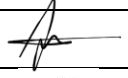


Assignment/Report Cover Sheet

Student Name	Student ID Number	Group No	Signature
ABDUL RAHIM BIN IDRIS	102368	GROUP 7	
PUTRA NURIMAN AKMAL BIN ROZIMAN	102917		
WAN ADAM DANISH BIN WAN ABIT	103014		
QHAIRUNNISA SHAHIRAH BINTI ZAKARIA	106089		

Subject Code : TMF2954	Subject Name : Java Programming
Assignment Title : Encoded	Lecturer : Madam Azima Alya
Due Date : 24 th May 2025	Date Submitted : 24 th May 2025

This cover sheet must be completed, signed and firmly attached to the front of the submission. All work must be submitted by the due date. If an extension of work is granted, an assignment extension acknowledgement slip must be signed by the lecturer/tutor and attached to assignment. Please note that it is your responsibility to retain copies of your assignment.

Plagiarism and Collusion are methods of cheating that falls under Peraturan Akademik Universiti Malaysia Sarawak para 11: Etika Akademik

Plagiarism

Plagiarism is the presentation of work which has been copied in whole or in part from another person's work, or from any other source such as the internet, published books or periodicals without due acknowledgement given in the text.

Collusion

Collusion is the presentation of work that is the result in whole or in part of unauthorized collaboration with another person or persons.

Where there are reasonable grounds for believing that cheating has occurred, the only action that may be taken when plagiarism or collusion is detected is for the staff member not to mark the item of work and to report or refer the matter to the Dean. This may result in work being disallowed and given a fail grade or if the circumstances warrant, the matter may be referred to a Committee of inquiry for investigation. Such investigation may result in the matter being referred to the University Discipline Committee, **which** has the power to exclude a student.

MARK :

Code Contribution Table

Member	Class/Component	Methods/Features	Responsibilities
QHAIRUNNISA SHAHIRAH BINTI ZAKARIA	Encoded class (core logic)	Package structure, class setup, countCharacters(), generateShift(), calculateFinalShift(), applyCipher(), getters	Overall encoding class structure; field initialization and core encoding logic and cipher operations
PUTRA NURIMAN AKMAL BIN ROZIMAN	Encoded class (logic)	checkStringValidity()	Input validation for lowercase letters, digits, and spaces; Empty input check (lines 170-184)
ABDUL RAHIM BIN IDRIS	Assignment and Encoded classes	initializeGUI(), setupActionListeners(), GUI components (lines 92-162)	GUI framework setup; JFrame, JTextField, JButton, JTextArea, JLabel initialization; UI component alignment and layout; Main method; Button action listener setup (partial). Collaborative implementation of button click logic and event handling (lines 166-211)
WAN ADAM DANISH BIN WAN ABIT	Assignment class (GUI)	charCountLabel (lines 126-128, 156), setupActionListeners(). UI error/success dialogs (lines 191-210)	Non-space character display label; Error and success message dialogs in action listeners; UI enhancement for result display

Group Shift Explanation

Group ID

```
class Encoded{

    private String inputText;
    private int charCount;
    private String resultText;
    private final String groupID = "G05/SE-G07"; // Hardcoded secret key
```

Figure 1 *Group ID*

Based on Figure 1, this is the official identification (“G05/SE-G07”) provided for this assignment; it was also kept as a constant in the code. In our program, we utilize it as a hidden internal value. Although the user cannot see it, it is necessary to create a distinct encoding pattern for others.

Group Shift

Based on the code, the groupShift is derived from line 52 in the generateShift() method:

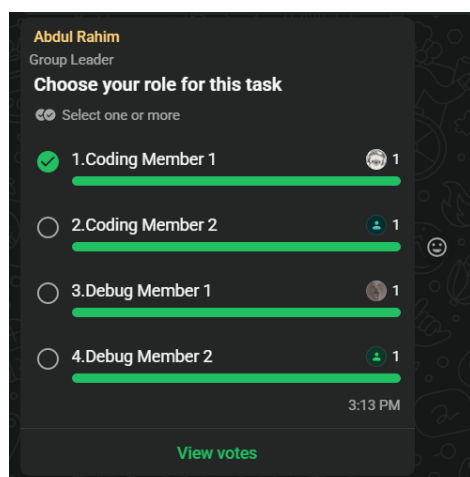
```
51 public int generateShift() {
52     return Math.abs(groupID.hashCode()) % 10 + 1;
53 }
```

Figure 2 *groupID.hashCode()*

The groupShift will be a value between 1 and 10 (inclusive), calculated from the embedded group ID “G05/SE-G07”.

Our group ID string will be transformed into a distinct integer using hashCode(). Java’s hashCode() converts any string into a 32-bit integer value using an algorithm based on the string characters. We also use absolute value because the number entered might be negative. Since the string “G05/SE-G07” never changes, the value is constant. It consistently returns the same number for the same string, Java’s hashCode() is deterministic. To turn the hash into a useful shift, we just use the fundamental mathematical operations (abs, %, +1). This implies that our program will generate the identical groupShift for our group and always between 1 and 10 each time it runs. Because each groupIDs will produce different hash values and shifts, this guarantees that the output is consistent for our group and that the encoding is distinct from that of other groups.

Proof of Discussion



Project Team	
Member Name	Responsibility
Member 1	[e.g., UI Design / Swing Implementation]
Member 2	[e.g., Encoding Logic / Hashing Mechanism]
Member 3	[e.g., Input Validation / Error Handling]
Member 4	[e.g., Documentation / Testing]
Member 5	[e.g., Repository Management / Code Integration]
Submission Deadline	
[cite_start]24th May 2026, 11:59 PM [cite: 3]	


JP SE G07
ColdShadow, Nisa, Putra, You



Nisa: salam hi guys, here's link for report n editing the code: • Link Report: https://unimas-my.sharepoint.com/:w:/g/personal/106089_si...



salam n hi guys, i added some code inside our repository. u guys can edit it / do corrections according to rubric if u want . tq @all

8:35 am

Aslm, everyone. Since our java requires all of us to do it. I have put the flow of each member who must do first.

Nisa ✓ -> Rahim(17 May) -> Wan(20 May) -> Putra(22 May)

10:36 pm ✓

Currently what you see here is the deadline for you to write, or edit the code before other members could start doing their part

10:37 pm ✓

Note that, you must do the following commands I sent for the GitHub so that other members could follow up with the latest code

10:38 pm ✓

PU Putra Unimas

just edit the same one that nisa created is it?

10:39 pm

Yes

10:41 pm ✓

Code
Blame
222 lines (179 loc) • 6.97 KB

Older
Newer

last week

106089, i have edited but some im...

1
package Assignment;
2
//JAVA PROGRAMMING ASSIGNMENT 2
3
//SE LG05 (G07)
4
5
//CONTRIBUTED BY NISA//
6
import javax.swing.*;
7

2 days ago

A few debugs along the way suc...

8
class Encoded{

last week

106089, i have edited but some im...

9
10
private String inputText;
11
private int charCount;
12
private String resultText;
13
private final String groupId = "G05/SE-G07"; // Hardcoded secret key
14

2 days ago

A few debugs along the way suc...

15
public Encoded() {
16
//Default constructor
17
}

last week

106089, i have edited but some im...

18

2 days ago

A few debugs along the way suc...

19
//Overloaded constructor for input processing and encoding
20
public Encoded(String inputText) {

last week

106089, i have edited but some im...

21
this.inputText = inputText;
22
this.charCount = countCharacters();